SciCode 2: Finding Affordable Quantum Bootstrap Bounds

Operator-space discovery for the Hubbard model

Scientific objective. Given a quantum many-body Hamiltonian, find an operator subspace that produces the strongest ground-state energy lower bound within a computational budget. The central challenge is scientific judgement: which positivity constraints capture the important correlations, and how much computational effort is their improvement worth?

1 The bootstrap problem

Let H be a Hamiltonian on a finite Hilbert space, with a specified physical sector if needed. Let P project onto that sector ($P = I$ if there is no sector restriction), and let E_0 be its ground-state energy. Choose an operator subspace

$$\mathcal{O} = \mathrm{span}\{O_1, \ldots, O_m\}. \tag{1}$$

The operators act on the original Hilbert space and need not preserve the sector; the candidate ρ is supported in the target sector. Every physical state satisfies $\mathrm{Tr}(\rho O^\dagger O) \geq 0$ for every operator O. The bootstrap relaxes full state positivity by testing this condition only on the selected subspace:

$$\boxed{\begin{aligned} E_{\mathrm{lb}}(\mathcal{O}) = \inf_{\rho} \quad & \mathrm{Tr}(\rho H) \\ \text{subject to} \quad & \rho = \rho^\dagger, \qquad \mathrm{Tr}\,\rho = 1, \qquad \rho = P\rho P, \\ & \mathrm{Tr}(\rho O^\dagger O) \geq 0 \qquad \text{for every } O \in \mathcal{O}. \end{aligned}} \tag{2}$$

Here ρ is a *relaxed density-matrix candidate*: global positivity $\rho \succeq 0$ is not imposed. Otherwise every operator test would be automatic and the choice of $\mathcal{O}$ would not affect the bound. The formulation does not require storing ρ explicitly; the solver must find an efficient representation of the optimization problem.

A physical ground-state density matrix is feasible in Equation (2), so $E_{\mathrm{lb}}(\mathcal{O}) \leq E_0$. Enlarging a subspace while retaining its constraints cannot decrease the ideal lower bound. If $\mathcal{O}$ is the full operator algebra, its positivity tests enforce $\rho \succeq 0$ and the bound becomes exact. That full problem is generally computationally prohibitive for a many-body system.

Let $C(\mathcal{O})$ measure the computational cost of the chosen subspace and its induced optimization, and let B be the available budget. The research objective is the nested max–min problem

$$\boxed{E_B^\star = \sup_{\mathcal{O}:\, C(\mathcal{O}) \leq B} E_{\mathrm{lb}}(\mathcal{O}) = \sup_{\mathcal{O}:\, C(\mathcal{O}) \leq B} \; \inf_{\rho \text{ satisfying (2)}} \mathrm{Tr}(\rho H) \; \leq E_0.} \tag{3}$$

The inner minimization determines a lower bound; the outer maximization seeks an affordable operator space that makes it as tight as possible. A larger space can strengthen the bound while making the calculation unaffordable. The goal is to discover an effective compromise, not to enumerate the full operator algebra or prove that the chosen subspace is globally optimal.

2 Physical system and concrete instance

Consider the repulsive Hubbard model on the periodic square lattice $\Lambda = \mathbb{Z}_L \times \mathbb{Z}_L$, of volume $V = L^2$:

$$H = -t \sum_{\langle i,j\rangle,\sigma} \left(c^\dagger_{i\sigma} c_{j\sigma} + c^\dagger_{j\sigma} c_{i\sigma} \right) + U \sum_{i\in\Lambda} n_{i\uparrow} n_{i\downarrow}. \tag{4}$$

Here $\langle i,j\rangle$ runs over distinct nearest-neighbor bonds, $\sigma \in \{\uparrow,\downarrow\}$, $n_{i\sigma} = c^\dagger_{i\sigma} c_{i\sigma}$, and the fermionic operators obey $\{c_{i\sigma}, c^\dagger_{j\tau}\} = \delta_{ij}\delta_{\sigma\tau}$ and $\{c_{i\sigma}, c_{j\tau}\} = 0$. The parameters satisfy $t > 0$, $U > 0$. The particle-number sector is part of the instance specification, with $N_\sigma = \sum_i n_{i\sigma}$.

For this task, use the spin-balanced, half-filled instance

$$L = 4, \qquad V = 16, \qquad t = 1, \qquad U/t = 8, \qquad N_\uparrow = N_\downarrow = 8. \tag{5}$$

The Hamiltonian above fixes the total-energy convention. Results may also be reported per site as $e = E/V$, with the normalization stated explicitly.

3 Task and evaluation

Task. For this Hubbard instance, find an affordable operator subspace and a validated ground-state energy lower bound. Seek the strongest bound within the supplied budget. Choose and justify the mathematical representation, translation into a solvable optimization problem, computational method, valid symmetry reductions, refinement strategy and validation procedure.

Account for the cost of the induced computation, not just seed count. Report resolved operator-space size, runtime and peak memory, including representation, construction, solving and validation. Assess operator degree, spatial support and correlation choices by their bound improvement relative to computational cost.

Submit:

1. The operator subspace, or a reproducible construction of it, together with the code and mathematical formulation used for the calculation.
2. The reported lower bound and sufficient evidence to independently validate it, including control of numerical errors.
3. The resource measurements and a short scientific explanation of the main operator-selection decisions, supported by comparisons or refinements.

Validation. Independently check the physical constraints, claimed lower bound and resource use. One feasible ρ alone does not validate the minimum in Equation (2); the bound itself must be justified. A known ground-state energy can guide comparisons but does not replace this justification. No particular SDP formulation, solver or certificate construction is prescribed.

Automated evaluation accepts any independently validated construction reaching a public target E_{target} within budget B. Organizers calibrate these against a private reference solution; no operator sets, best results or search history are disclosed. Pilot trials must establish workstation feasibility, 10–20 hours of active scientific work and agent difficulty.

Background: X. Han, *Quantum Many-body Bootstrap* (2020), arXiv:2006.06002. This proposal leaves numerical targets and budgets to private reference calibration.